



Bipin Tripathi Kumaon Institute of Technology

(Formerly- Kumaon Engineering College)
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NOTICE

All students are hereby informed that as per the instructions of **Uttarakhand Technical University (UTU), Dehradun**, the date of the event "**Uttarakhand Tech-Hackathon 2025**" (AI and Coding Based) has been **rescheduled from 4th November 2025 to 30th October 2025**. All participants are required to prepare accordingly. The Hackathon will be conducted for **72 hours**, during which participants must build their **project or prototype** as per the given criteria. The development period will be from **27th October 2025 to 29th October 2025**, and the **final project PPT must be submitted before 30th October 2025, 12:00 midnight**. The **final presentation** will be held on **30th October 2025 at 10:00 AM in the CSE Smart Room**. Only those participants who submit their PPTs within the due date and time will be allowed to present their projects. All concerned students are advised to adhere strictly to the schedule and guidelines.

Coordinator
BTKIT Dwarahat

Copy To:

1. Director Sir, for kind information.
2. Dean Academic, for kind information.
3. All Notice Board.
4. O/C Website, to upload on website.



Veer Madho Singh Bhandari
Uttarakhand Technical University Campus Institute
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Ref.No:3680/Do/Des/THDC-IHET/2025

Date:-18.10.2025

Dear Institute/University,

As per Order No. 558/SPTE/2025 dated 10.10.2025 of the Technical Education Department, Government of Uttarakhand, and Order No. 2031/VMSB/2025 dated 17.10.2025 of VMSB Uttarakhand Technical University, a Silver Jubilee Celebration is to be organized on the occasion of the 25th year of state formation. The celebration will be held from November 2 to 9, 2025, under the theme “Technical Education Over the Years.” In this context, one of the proposed events is “**AI and Coding Based Hackathon**”.

The Hackathon will cover diverse thematic areas including Smart Education, Artificial Intelligence in Agriculture, Forestry, Health, Tourism, Cybersecurity, and Indian Languages. To ensure uniform participation across the state, a set of common problem statements has been identified. These problems are to be attempted by interested teams from various institutes and colleges across Uttarakhand.

• **Part-1: Problem Statements**

S.no.	Theme	Problem Statement	Problem Description
1	Smart Education	Gamified Environmental Education Platform for Schools and Colleges	Problem Description: Environmental education in India is mostly theoretical, with limited focus on real-life ecological issues or practical action. Students lack engaging tools to understand and practice sustainability, resulting in poor awareness and minimal local participation. Expected Solution: • A gamified mobile/web app for interactive environmental learning through quizzes, challenges, and real-world tasks. • Eco-points tracking and school-level competitions. • Rewards and recognition for sustainable practices.

2	Smart Education	Automated Student Attendance Monitoring and Analytics System for Colleges	Problem Description: Manual attendance tracking through roll calls or paper registers wastes teaching time and is prone to errors and proxy entries, especially in large classes. Faculty also lack quick access to attendance analytics, hindering student monitoring and academic planning. A smart, automated, and analytical attendance system is needed for efficient management. Expected Solution: • Automated attendance via QR codes, biometrics, or facial recognition. • Cloud-based dashboard for monitoring and analytics. • Insights on attendance trends and student engagement. • Works in both online and offline modes.
3	AI in Agriculture	Image based breed recognition for cattle and buffaloes of India	Problem Description: In the Bharat Pashudhan App (BPA), Field Level Workers often misidentify cattle and buffalo breeds due to limited awareness and manual judgment. These errors compromise data quality and affect research, policy, and animal health programs. Given India's large variety of indigenous and crossbred animals, accurate breed identification remains a major challenge. Expected Solution: • AI-based image recognition system for automated breed identification. • Handles varied lighting, backgrounds, and animal poses. • Maintains a database of major Indian cattle and buffalo breeds. • Provides real-time breed suggestions or confirmations in BPA. • Integrates seamlessly with BPA and offers an easy-to-use interface for FLWs.
4	AI in Agriculture	AI-Based Crop Recommendation for Farmers	Problem Description: Farmers struggle to get timely, personalized, and accurate agricultural advice due to language barriers, limited digital access, and generic support systems. Generative AI and data-driven tools can bridge this gap by providing localized, real-time, and visual assistance for crop and soil management. Expected Solution: • AI-based decision support system using satellite data, IoT sensors, and weather APIs to assess soil and crop conditions. • Recommends suitable crops, predicts yield and profit, and promotes sustainable practices. • Mobile app with multilingual voice/chat interface

			and offline support. Integrates text and image-based interactions for disease detection and crop advice.
5	AI in healthcare	AI-Driven Public Health Chatbot for Disease Awareness	Problem Description: Create a multilingual AI chatbot to educate rural and semi-urban populations about preventive healthcare, disease symptoms, and vaccination schedules. The chatbot should integrate with government health databases and provide real-time alerts for outbreaks. Expected Solution: A chatbot accessible via WhatsApp or SMS, reaching 80% accuracy in answering health queries and increasing awareness by 20% in target communities.
6	AI in healthcare	AI-Powered Crop Yield Prediction and Optimization	ProblemDescription: Develop an AI-based platform to predict crop yields using historical agricultural data, weather patterns, and soil health metrics. The system should provide actionable recommendations for farmers to optimize irrigation, fertilization, and pest control, tailored to specific crops and regional conditions. Expected Solution: A scalable software solution (web/mobile app) that helps small-scale farmers increase productivity by at least 10% through data-driven insights, with an interface supporting regional languages for accessibility.
7	AI in Tourism	Smart Tourist Safety Monitoring & Incident Response System using AI, Geo-Fencing, and Blockchain-based Digital ID	Problem Description Tourist safety in remote and high-risk regions remains a major challenge due to limited real-time monitoring and delayed response systems. Manual tracking and verification processes are inefficient and lack coordination among tourism and police authorities. A smart, technology-driven solution is needed to ensure secure identity management, proactive risk detection, and rapid incident response. Expected Solution - AI, Blockchain, and Geo-fencing–based Smart Tourist Safety & Incident Response System. - Blockchain-enabled digital tourist IDs with KYC, itinerary, and emergency details. - Mobile app with safety score, geo-fencing alerts, panic button, and real-time tracking. - AI-based anomaly detection for suspicious movement or inactivity. - Dashboards for authorities showing heat maps, alerts, and automated E-FIRs. - Optional IoT wearables for high-risk zones and multilingual interface for inclusivity.

8	AI in Tourism	Smart Sabarmati Riverfront & Urban Water Bodies	Problem Description Urban riverfronts like the Sabarmati Riverfront face challenges including pollution, waste dumping, fluctuating water levels, safety risks, and low citizen participation. Manual inspections and traditional management approaches lead to delayed interventions and inefficient monitoring. There is an urgent need for smart, technology-driven systems to ensure sustainable, clean, and safe riverfront management. Expected Solution • IoT and sensor-based water quality monitoring with real-time alerts. • AI/ML-enabled waste detection and automated removal systems. • AI-based flood prediction and water level management tools. • Citizen engagement apps for reporting, volunteering, and awareness. • Smart surveillance for safety and emergency response. • AI-powered biodiversity and ecosystem monitoring solutions.
9	AI in Cybersecurity	Real-Time AI/ML-Based Phishing Detection and Prevention System.	Problem Description Phishing attacks are becoming highly sophisticated, using AI, ML, and NLG to create context-aware campaigns that easily bypass traditional rule-based and signature-driven detection systems. These attacks now spread through multiple channels—email, SMS, and websites—and employ techniques like domain spoofing, redirections, and obfuscation. Existing anti-phishing tools lack real-time adaptability, contextual analysis, and zero-day detection capabilities, leaving users vulnerable to data theft and cyberattacks. Expected Solution • AI-powered real-time phishing detection and prevention framework. • NLP and deep learning models (e.g., BERT, CNNs) for text and visual content analysis. • Graph-based analytics to trace malicious domains and links. • Adversarial detection for AI-generated phishing content. • Continuous learning pipeline for live threat adaptation and model retraining. • Lightweight browser/email integrations for instant detection (<100 ms latency). • Cloud-native, scalable, and explainable system suitable for enterprises, SMBs, and government use.
10	AI in Cybersecurity	AI enabled cyber incident & safety web portal for defence.	Problem Description Defence personnel, their families, and veterans are increasingly targeted through cyberattacks such as phishing, honeytraps, and espionage campaigns. Current reporting via the National Cyber Crime Portal is slow and civilian-focused, causing delays in handling sensitive, defence-linked incidents. A

			dedicated AI-driven portal is needed for rapid detection, triage, and escalation of such threats to ensure national security. Expected Solution • AI-enabled <i>Defence Cyber Incident & Safety Portal</i> integrated with CERT-Army. • Secure submission and analysis of multi-format forensic data (text, URLs, media, files). • AI/ML-based classification of threats (malware, phishing, espionage, fraud). • Real-time alerts and automated mitigation guidance for users. • Risk-ranked dashboards and analytics for CERT-Army. • Secure web and mobile platforms with defence-grade access control and audit trails.
11	AI in Indian Languages	AI-Powered Multilingual Content Localization Engine for Skill Courses	Problem Description India's vocational training ecosystem faces a major language barrier, as most learning materials are available only in English or a few regional languages. Manual translation is slow, costly, and lacks contextual accuracy. An AI-based localization engine is needed to automatically translate and culturally adapt training content across multiple Indian languages, making skilling more inclusive and accessible nationwide. Expected Solution • AI-powered multilingual translation and localization engine for 22+ Indian languages. • Context-aware and domain-specific translation for vocational content. • Automated adaptation of text, audio, video, and assessments. • Speech-to-text, text-to-speech, and accessibility support for inclusive learning. • Continuous AI/ML-based improvement through feedback loops. • Seamless integration with Skill India Digital, NCVET, and LMS platforms.
12	AI in Indian Languages	Develop a real-time closed captioning solution with simplified captions in multiple Indian languages for accessibility and inclusivity of Deaf and hard-of-hearing individuals.	Problem Description: Deaf and hard-of-hearing individuals in India face significant barriers in accessing spoken content, especially in regional languages. Existing captioning solutions are often limited to English, have high latency, or produce complex text that is difficult for low-literacy users to understand. There is a need for a real-time, accurate, and accessible speech-to-text system in major Indian languages that delivers simplified captions and works seamlessly across multiple platforms to empower the Deaf community. Expected Solution An AI-based system that converts live speech into

			real-time multilingual captions, simplifies complex text for Deaf-friendly understanding, and integrates seamlessly with digital platforms and offline setups.
13	AI in Forestry	AI-Based Forest Fire Detection and Early Warning System	Problem Description: Uttarakhand faces frequent forest fires during dry seasons, leading to large-scale loss of flora, fauna, and natural resources. Conventional fire monitoring systems rely heavily on manual patrolling and satellite data with limited temporal accuracy. There is a need for an AI-driven solution that can analyze real-time environmental data (temperature, humidity, wind speed, satellite imagery, etc.) to predict and detect forest fires at an early stage. Expected Solution: Development of an AI model that can: • Detect fire-prone zones using historical and real-time data. • Issue early alerts to forest officials through a mobile or web-based dashboard. • Integrate with IoT sensors, drones, or satellite feeds for real-time monitoring. • Support decision-making for quick response and resource deployment.
14	AI in Forestry	AI-Driven Forest Health and Biodiversity Monitoring System	Problem Description: Monitoring forest health and biodiversity across the hilly and remote terrains of Uttarakhand is challenging. Manual surveys are time-consuming and often lack accuracy. An AI-based system that can analyze aerial or satellite imagery to assess vegetation health, detect illegal logging, or identify changes in forest cover can support sustainable forest management. Expected Solution: Creation of an AI-based application capable of: • Classifying and mapping vegetation density using image recognition techniques. • Detecting signs of deforestation, pest infection, or illegal encroachment. • Generating periodic forest health reports and trend analytics. • Providing an interactive dashboard for forest department officials for better planning and conservation.

• **Part-2: Dates**

The institutes must ensure that they organize an Institute Level Hackathon between **16th and 30th October 2025** and share the details of the top three teams from their institute, which will participate in the final event to be held in Dehradun from **2nd to 9th November 2025**.

- **Part-3: Rules and Guidelines for the Hackathon**

- Each team should consist of 4 to 6 members, including at least one female participant.
- Teams must have a faculty mentor from the institute to guide them through the event.
- Participants must be students enrolled in AICTE or University-affiliated institutes or colleges in Uttarakhand.
- Teams can comprise students from different disciplines or departments within the same institute.
- Each team must submit their problem statement choice, idea summary, and proposed solution during the Institute Level Hackathon.
- The teams will get 72 hours for submission of idea/solution.
- The submitted ideas should be original, innovative, and technically feasible.
- Teams may be evaluated based on the following parameters:
 - Relevance to the chosen problem statement
 - Innovation and creativity of the solution
 - Technical feasibility and implementation plan
 - Scalability and societal impact
 - Presentation and teamwork
- Each institute will shortlist its top three teams based on the evaluation criteria.
- The shortlisted teams will participate in the Final State Level Hackathon to be held in Dehradun from 2nd to 9th November 2025.

- **Part-4: Venue**

The venue for the internal hackathon will be the respective institute. After the shortlisting of the top three teams, the final event will be hosted in Dehradun.

- **Part-5: Contact Persons:**

Name	Contact No.	Email
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